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# Sample-Efficient Bayesian Process Optimisation with Structure-Aligned, Prior-Informed Deep Gaussian Processes

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## Introduction

Optimising chemical process plants remains costly and time-consuming, often demanding iterative changes to full-scale plants despite the availability of rigorous models and lab-scale set-ups. In recent years, machine learning, particularly **Bayesian optimisation** (BO) [1] with Gaussian process (GP) surrogates [2], has emerged as a compelling approach for expensive or noisy objectives, offering principled uncertainty quantification and high sample efficiency [3]. With suitable kernels and informative priors, GPs can embed domain knowledge and perform well in the low-data regime [4]. **Deep Gaussian Processes** (DPGs) extend GPs by composing layers akin to multilayer perceptions (MLPs), enabling richer, non-Gaussian predictive behaviour and better modelling of hierarchical dependencies intrinsic to chemical processes [5]. Recent advances in variational training focus on rendering DGPs computationally tractable in the high-data regime [6], making them theoretically viable surrogates for high-throughput BO of expensive targets. Yet, training DGPs on small datasets remains a challenge [7].

## Methodology

This work presents a proof of concept for a **structure-aligned Deep Gaussian Process (DGP)** integrated within a Bayesian optimisation routine. The framework is specifically designed for multi-step (also named composite) target functions where lower-fidelity (LF) data is available for individual process steps - a common scenario during process scale-up. Unlike standard "black-box" surrogates, our model architecture mirrors the physical topology of the chemical process. Features are not solely fed into the input layer but can be injected at intermediate layers, reflecting how specific control parameters influence distinct downstream process steps.

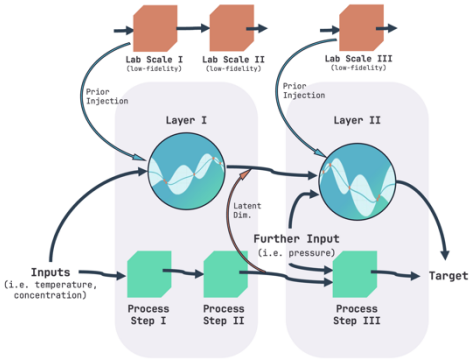



Fig. 1: Example of a structure-aligned DGP with two layers. The first layer encompasses the first two steps of the process.

To guide the model in data-sparse environments, we employ a residual learning strategy. Individual GPs are pre-trained on LF data for each process step, and their posterior means serve as the mean functions for the corresponding DGP layers. Consequently, the DGP learns the deviation between the idealised LF physics and the true high-fidelity observations. For rigorous uncertainty quantification, we leave variational methods out of consideration in favour of fully Bayesian inference via **No-U-Turn Sampling** (NUTS), a robust Hamiltonian Monte Carlo (HMC) method.

## Implementation

We employ a hybrid JAX/PyTorch framework in Python that balances rigorous sampling with practical usability. Parameter inference is executed using NUTS in **NumPyro (JAX)**, leveraging XLA compilation and chain vectorisation for efficiency. The resulting posterior samples are transferred to a **PyTorch/BoTorch** environment for acquisition function optimisation. To guarantee numerical

consistency and stability, all kernels, distance metrics, and inference routines were explicitly implemented with identical logic across both backends, ensuring that the JAX-based sampling maps exactly to the PyTorch-based decision-making.

### Experiments

The framework was evaluated on two custom multi-step benchmarks derived from first-principles engineering: a three-step **discontinuous polymerisation process** (3D input) and a four-step **complex pharmaceutical flow sheet** (5D input). These benchmarks utilise algebraic formulations to ensure fast runtime while mimicking the hierarchical dependencies of real systems. For both cases, we employed a **two-layer structure-aligned DGP architecture** tailored to the process topology. In the polymerisation benchmark, the first layer processes all inputs to generate two latent variables for the output layer. The pharmaceutical benchmark, however, requires a more complex connectivity: specific inputs are **injected** directly into the second layer, interacting with three latent variables propagated from the first layer to accurately model the downstream process steps. We compared this structure-aligned approach against a standard SingleTaskGP (exact GP) BO routine over multiple repetitions. **Early results indicate** that the structure-aligned DGP, whilst computationally more demanding, **achieves superior sample efficiency** and optimisation performance within fewer iterations (see **Fig 1.**).

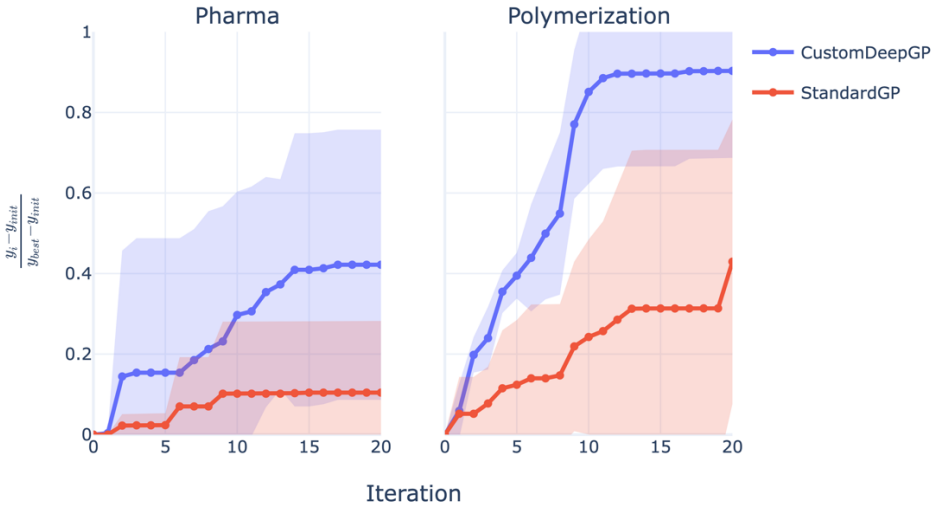



Fig. 2: Normalised Improvement $\left(\frac{y_i - y_{init}}{y_{best} - y_{init}}\right)$ of the structure-aligned DGP (CustomGP) and a SingleTaskGP (StandardGP) within a BO routine leveraging Log Expected Improvement on both benchmarks. Five repetitions each. Visualised are the according means and standard deviations.

### Outlook

Future work will expand the empirical evidence base through more extensive benchmarking and longer optimisation runs. Additionally, we aim to benchmark this approach against other advanced surrogates, such as warped GPs, to further validate the benefits of structural alignment in process optimisation. Nevertheless, this work highlights a successful proof of concept for structure-aware Bayesian optimisation in chemical engineering.

## References


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